

Hotel Prices vs. Location Analysis Report

Course:

Data Collection and Preprocessing

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Task Distribution

Name and ID	Code File	Final Report	Contribution
Hala Alrefe | 446016956	• Outliers Detection and Handling • Analysis & Results • Text Preprocessing	• Conclusion • Limitations	20%
Mayar Matook | 446003113	• Data Quality Assessment • Data Cleaning & EDA	• Data Quality Assessment	20%
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Ritaj Alhosawi | 446007287	• Data Loading & Inspection	• Introduction & Data Overview	20%

1. Introduction & Data Overview

This project analyzes a hotel dataset from the Netherlands to explore hotel ratings, reviews, locations, and accommodation types. The main goal is to apply data preprocessing techniques, identify and resolve data quality issues, perform exploratory data analysis (EDA), and discover meaningful patterns within the dataset. Through cleaning, transformation, visualization, and statistical analysis, the project provides insights into hotel characteristics and customer experience across different cities.

Dataset Description:

The dataset contains information about hotels in the Netherlands. It includes **525 rows and 9 columns**, covering hotel prices, ratings, review counts, locations, and other descriptive hotel attributes.

Dataset Source:

The dataset was obtained from **Kaggle** and contains hotel-related information collected from different cities in the Netherlands. Hotel Booking Dataset

The Dataset Represents:

The dataset represents characteristics of hotels across multiple Dutch cities, including numerical and categorical features related to hotel quality and customer experience such as hotel names, prices, review counts, ratings, and locations.

Objective of the Analysis:

The objective of this project is to perform a complete data preprocessing pipeline, identify and fix data quality issues, apply feature engineering and feature extraction techniques, and discover useful insights related to hotel prices and ratings.

Why This Dataset?

This dataset was selected because it contains realistic data quality problems such as missing values, duplicates, and outliers, making it suitable for practicing preprocessing techniques and generating meaningful analytical insights.

2. Data Quality Assessment

Missing Values

Missing values were detected using: **df.isnull().sum()**

Missing values may produce incomplete analysis results and can affect statistical calculations and visualizations, leading to inaccurate conclusions.

Duplicate Records

Duplicate rows were identified using: **df.duplicated().sum()**

Duplicate records may create bias because the same observation is counted multiple times, which can affect summary statistics and visualizations.

Outliers

Outliers were detected using **Boxplot visualization and the IQR rule.**

Outliers may affect statistical measures such as mean values and may lead to misleading analysis results.

Detect Outliers using IQR Rule

In [27]:

```
Q1 = df[column].quantile(0.25)
Q3 = df[column].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

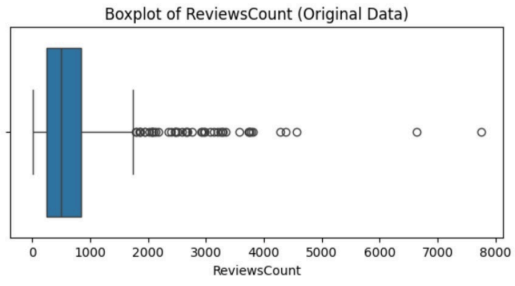
print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower bound:", lower)
print("Upper bound:", upper)

outliers = df[(df[column] < lower) | (df[column] > upper)]
print("Number of Outliers", len(outliers))
```

```
Q1: 244.0
Q3: 841.0
IQR: 597.0
Lower bound: -651.5
Upper bound: 1736.5
Number of Outliers 47
```

In [26]:

```
plt.figure(figsize=(7,3))
sns.boxplot(x=df[column])
plt.title("Boxplot of ReviewsCount (Original Data)")
plt.show()
```



Inconsistent Labels

Inconsistent labels were checked using :

df['City'].value_counts()

df['Type'].value_counts()

The frequency of categories was examined to identify possible inconsistencies in labels.

Inconsistent labels can split the same category into multiple groups, which may affect grouping results and produce inaccurate summaries and visualizations.

Invalid Values

Invalid values were checked by searching for impossible values such as:

df[df['Hotel_Price'] < 0]

df[(df['Rating'] < 0) | (df['Rating'] > 5)]

df[df['ReviewsCount'] < 0]

Invalid values represent incorrect information and may affect the reliability and accuracy of the analysis.

```
Invalid negative prices:
Empty DataFrame
Columns: [Unnamed: 0, Name, Place, Type, Hotel_Price, ReviewsCount, Ra
ting, City, State]
Index: []
Invalid ratings:
     Unnamed: 0                                Name  \
0             0               BUNK Hotel Amsterdam
1             1                    YOTEL Amsterdam
2             2                    Multatuli Hotel
3             3                 nhow Amsterdam Rai
4             4                Motel One Amsterdam
..          ...                                ...
520         520    Stadslogement Bij de Sassenpoort
521         521               Mercure Hotel Zwolle
522         522       The Cabin at Zwolle Centraal
523         523                  Hanze Hotel Zwolle
524         524  Campanile Hotel & Restaurant Zwolle

[523 rows x 9 columns]
Invalid negative review counts:
Empty DataFrame
Columns: [Unnamed: 0, Name, Place, Type, Hotel_Price, ReviewsCount, Ra
ting, City, State]
Index: []
```

Mixed or Incorrect Data Types

Data types were checked using: **df.dtypes**

Incorrect data types can cause errors in calculations and prevent proper analysis and visualization.

3. Data Cleaning Process

Several preprocessing steps were applied to clean the dataset and improve its quality before analysis.

Handling Missing Values

Missing values were identified using:

df.isnull().sum()

The dataset contained missing values in several columns, especially:

Hotel_Price,State,Rating,ReviewsCount

Missing values in numerical columns such as Rating and ReviewsCount were handled using median imputation to reduce the effect of extreme values and preserve the dataset size.

Removing Duplicate Records

Duplicate rows were checked using:

df.duplicated().sum()

Any duplicated records were removed to prevent repeated observations from affecting

the analysis and visualizations.

Handling Outliers

Outliers were detected using:

- Boxplot visualization
- IQR (Interquartile Range) method

Extreme values in the ReviewsCount column were identified and removed because they could negatively affect statistical analysis and correlation results.

Fixing Inconsistent Labels

Categorical columns such as City and Type were checked for inconsistent labels using:

df['City'].value_counts()

df['Type'].value_counts()

Label consistency was improved by correcting formatting issues and ensuring categories were standardized.

Handling Invalid Values

The dataset was checked for invalid numerical values using conditions such as:

df[df['Hotel_Price'] < 0]

df[(df['Rating'] < 0) | (df['Rating'] > 10)]

df[df['ReviewsCount'] < 0]

Invalid values were corrected or removed to maintain data reliability.

Correcting Data Types

Data types were verified using: **df.dtypes**

Numerical columns were converted into appropriate numeric formats to support calculations and analysis.

4. Data Transformation & Feature Engineering

Several transformation and feature engineering techniques were applied to prepare the dataset for analysis and improve data interpretation.

Encoding

Categorical variables were transformed into numerical representations when necessary to make the dataset compatible with analysis techniques and machine learning algorithms.

Scaling

Numerical features were scaled to reduce differences between variable ranges and improve analysis consistency. Scaling helps prevent variables with large values from dominating the analysis.

Feature Engineering

New features were created to improve understanding of the dataset and simplify comparisons between hotels.

The following new features were generated:

- Rating_Category
- Reviews_Level

These features grouped hotels into meaningful categories based on customer ratings and review counts.

PCA (Principal Component Analysis)

PCA was applied to reduce dimensionality and simplify the dataset structure while preserving the most important information. This helped improve visualization and analytical efficiency.

Benefits of Transformation

The transformation process improved:

- Dataset consistency
- Feature usability
- Analysis quality
- Interpretation of hotel patterns and customer behavior

Feature engineering also helped reveal clearer relationships between hotel ratings, reviews, and locations.

5. Analysis Results

Final Summary of the Cleaned Dataset

After preprocessing, the dataset became cleaner and more suitable for analysis. Missing values were handled using mean/median techniques, duplicate records were removed, and outliers in **ReviewsCount** were detected and removed using the IQR method.

Detect Outliers using IQR Rule

In [27]:

```
Q1 = df[column].quantile(0.25)
Q3 = df[column].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower bound:", lower)
print("Upper bound:", upper)

outliers = df[(df[column] < lower) | (df[column] > upper)]
print("Number of Outliers", len(outliers))
```

```
Q1: 244.0
Q3: 841.0
IQR: 597.0
Lower bound: -651.5
Upper bound: 1736.5
Number of Outliers 47
```

The original dataset contained **525 rows and 9 columns**. New features were created during feature engineering:

- Rating_Category
- Reviews_Level

PCA was also applied to simplify the data structure while preserving useful information.

Descriptive Statistics of Key Variables

The main variables used for analysis were:

- Rating
- ReviewsCount
- Hotel_Price

The ratings mostly ranged between **7 and 10**.

```
     Hotel_Price  ReviewsCount  Rating                  City
State
0            NaN         778.0     8.4        Amsterdam Noord   A
msterdam
1            NaN         500.0     8.1        Amsterdam Noord   A
msterdam
2            NaN        1605.0     7.4  Amsterdam City Center   A
msterdam
3            NaN         500.0     9.0           Zuideramstel   A
msterdam
4            NaN         500.0     8.8           Zuideramstel   A
msterdam
..           ...           ...     ...                    ...
...
520          NaN         232.0     8.4                Zwolle   A
msterdam
521          NaN        1402.0     7.7                Zwolle   A
msterdam
522          NaN         149.0     7.2                Zwolle   A
msterdam
523          NaN        1095.0     7.2                Zwolle   A
msterdam
524          NaN        2071.0     6.4                Zwolle   A
msterdam
```

Review counts varied significantly between hotels. Some hotels had extremely high review counts which created outliers.

The correlation value between **Rating** and **ReviewsCount** was:

−0.143

which indicates a weak negative relationship.

Visualization Interpretation

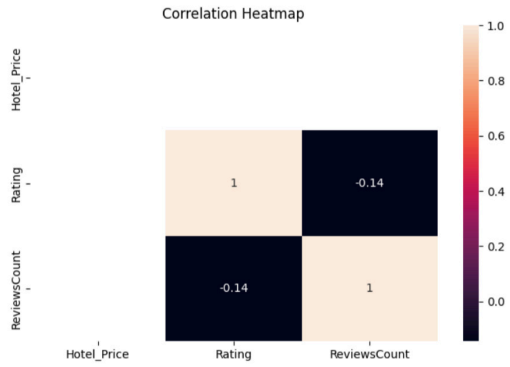
Two main visualizations were used:

1- Correlation Heatmap

The heatmap showed a weak negative relationship between **Rating** and **ReviewsCount**.

This means hotels with a larger number of reviews did not necessarily receive higher ratings.

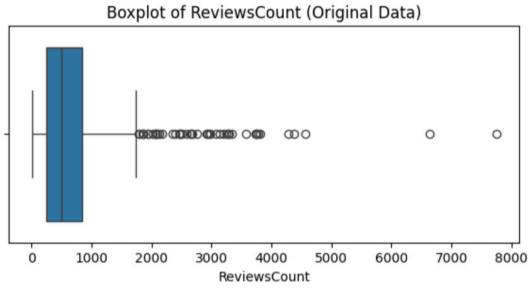
In [24]:
```
plt.figure(figsize=(8,5))
sns.heatmap(df[['Hotel_Price', 'Rating', 'ReviewsCount']].corr(),
annot=True)
plt.title('Correlation Heatmap')
plt.show()
```



2- Boxplot for ReviewsCount

The boxplot identified several extreme values in the **ReviewsCount** column.

In [26]:
```
plt.figure(figsize=(7,3))
sns.boxplot(x=df[column])
plt.title("Boxplot of ReviewsCount (Original Data)")
plt.show()
```



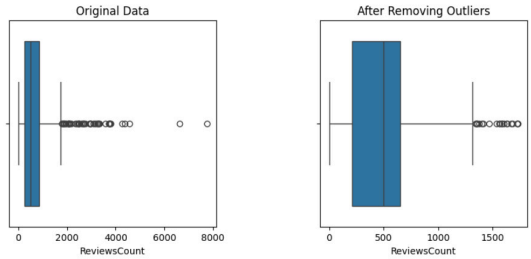
After removing these outliers, the distribution became more balanced and suitable for analysis.

Compare Before & After

In [29]:
```
fig, ax = plt.subplots(1, 2, figsize=(10,4))
plt.subplots_adjust(wspace=0.5)

sns.boxplot(x=df[column], ax=ax[0])
ax[0].set_title("Original Data")

sns.boxplot(x=df_removed[column], ax=ax[1])
ax[1].set_title("After Removing Outliers")
plt.show()
```



Important Patterns Discovered

The analysis revealed several patterns:

- Hotels with high review counts did not always have high ratings.
- Review count alone was not a strong indicator of hotel quality.
- Hotels were concentrated mainly in larger cities such as Amsterdam.
- Feature engineering made hotel groups easier to compare.

Relationships Between Variables

The relationship between **Rating** and **ReviewsCount** was weak according to the correlation value:

−0.143

This indicates that increasing the number of reviews does not strongly affect hotel ratings.

Key Insights Supported by Data

1. Customer ratings depend on factors beyond review quantity.
2. Feature engineering improved the understanding of the dataset by creating meaningful categories.
3. Removing outliers improved dataset quality and reduced the impact of extreme values.
4. PCA simplified the dataset while preserving useful information.
5. Data preprocessing improved overall analysis performance and made the dataset more reliable.

Conclusion

Our dataset shows that hotel listings across the Netherlands are generally well-rated, with an average rating of 8.33 out of 10 and ratings ranging from 3.7 to 10.0. Double Room is the most dominant accommodation type in the dataset, indicating it is the most widely offered option across cities. Delfshaven stands out as the highest-rated city with an average rating of 9.10, notably above the overall average. Our dataset also shows high variability in ReviewsCount (1 to 7748), suggesting that some listings are significantly more popular and reviewed than others. The weak negative correlation (−0.14) between Rating and ReviewsCount indicates that higher ratings do not necessarily come with more reviews.

Limitations

The Hotel_Price column had 525 missing values (all entries null), making any price-related analysis impossible, this is a major limitation in our dataset.

- The State column was missing 392 values (~75%), which limited state-level analysis.
- Missing values in Rating and ReviewsCount were filled using median imputation, which may introduce slight **bias** and not reflect true values.
- The dataset covers only 525 records from selected Dutch cities, which may not represent all hotels across the Netherlands.
- 47 outliers in ReviewsCount were detected and handled via removal and log transformation; however, some genuine high-traffic hotels may have been affected.
- Future analysis would benefit from a larger dataset with complete price data and broader geographic coverage.